

a Unified Output Adaptation

Refer to:

"the woman in red holding the umbrella"

Soft probability

Intervention field



$$u = 1 - |2p - 1|$$

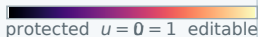
Hard mask

Boundary permission u



$$u = e^{-d(x, \partial M)^2 / (2\sigma^2)}$$

Unified black-box interface
 $(p, u) = \mathcal{A}(O)$



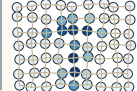
b Reliability-Weighted Regional

Semantic evidence - what to preserve

Pool p over SLIC

Pool u over SLIC

Region graph



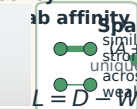
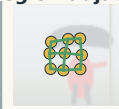
targets t_i

- confident FG strong anchor
- confident BG strong anchor
- uncertain
- weak anchor

$$a_i = (1 - u_i) + \epsilon_a + K_{\text{seen}}$$

Image structure - where to propagate

SLIC superpixels



ab affinity W_{ij}

- similar
- strong
- unique, range-preserving across edges
- weak

Sparse SPD solve

$$(A + \delta I)q^* = At$$

Regional field q



c Oriented Pixel R

Original

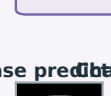
Intervention

Lift q^* or



Fusion - pixel-wise gate

$$\alpha = u^\beta, \quad \hat{p} = (1 - \alpha)p + \alpha r$$



Base prediction

Intervened

Ref result

Localized contour update



original
refined

$$|p(x) - p(x)| \leq u(x)^\beta$$

OUTPUT ONLY • NO INTERNAL FEATURES • NO GRADIENTS • NO LEARNED MASK DECODER